

ANTIDIABETIC POTENTIAL OF HYDRO-ALCOHOLIC EXTRACT OF *MORINGA PEREGRINA* LEAVES: IMPLICATION AS FUNCTIONAL FOOD FOR PROPHYLACTIC INTERVENTION IN PREDIABETIC STAGE

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ABSTRACT

In recent years, the herbal and dietary sources of bioactive components having pharmacological properties and therapeutic significance in traditional system of medicine have attracted much attention in complementary and adjuvant therapies. Some preclinical studies on animal models have shown benefits of *Moringa* species such as *M. oleifera* in experimentally induced diabetes. Most of these studies have taken the anti-hyperglycemic effect as the end-point indicator of antidiabetic properties. The present study has explored the antidiabetic potential of *M. peregrina*, another species of *Moringa*, native to Africa and Arabian Peninsula. The hydro-alcoholic (methanol-aqueous) extract of the dried leaves of the plant demonstrated inhibitory activity against three *in vitro* models of enzyme assay (α -amylase, α -glucosidase and dipeptidyl peptidase IV) critical for diabetes management. The antioxidant property was also evaluated which correlated with the phytochemical analysis of the extracts displaying the presence of phenolics, tannins and saponins. These findings provide partial evidence to support the traditional use of *Moringa* in diabetes and endorse its regular consumption as functional food for high-risk populations at the borderline prediabetic stage.

PRACTICAL APPLICATIONS

M. peregrina is commonly consumed as food and serves as an important ingredient in traditional medicine in Arabian peninsula. The assessment of the antidiabetic potential of the plant provides an insight into its application as functional food and in nutraceutical development for prophylactic intervention in hyperglycemic conditions.

INTRODUCTION

According to the statistics of International Diabetes Federation (IDF), currently the global burden of diabetes is estimated to be 382 million. A further 316 million with impaired glucose tolerance is at high risk from the disease – an alarming number that is set to reach 471 million by 2035 (IDF 2013). The report also presented diabetes as a cause of 5.1 million deaths until the end of 2013. Such a projection demands a proactive approach in the form of prophylactic interventions to avoid the incidence of active disease in prediabetic population. In congruence with this idea, it is thus

believed that diabetes control may benefit from the antidiabetic potential of alternative therapies that may include use of natural products of herbal/dietary origin which nonetheless have been an effective curative strategy reported for a number of diseases since ancient times (Saraswat *et al.* 2009). It has been reported that about 75% of U.S. households use dietary approaches to reduce their risk of chronic diseases, a philosophy widely accepted and practiced in countries like India and China for centuries (Sloan 2005; Christine and Milner 2011). The currently used conventional therapies for diabetes include stimulation of endogenous insulin secretion, enhancement of the action of

insulin at the target tissues, oral hypoglycemic agents such as biguanides and sulfonylureas and the inhibition of degradation of dietary starch by targeting intestinal glycosidases such as α -amylase and α -glucosidase (Leena and Jill 2010). The treatment regimen of type 2 diabetes mellitus worldwide has prescribed the use of metformin and sulfonylurea in combination as standard first-line antidiabetic therapy for years. However, long-term use of these drugs results in a progressive decline in β -cell function, and by 3 years up to 50% of diabetic patients need an additional pharmacological agent to maintain the glycosylated hemoglobin (HbA1c) below 7.0% (Gupta and Kalra 2011). Therefore, it is understood that the clinical value of traditional medicine which is also part of human diet and possesses higher safety index should be exploited to avoid the limitations associated with the long-term use of current therapies, particularly for prediabetic population. *Moringa* species belonging to monogeneric family Moringaceae, particularly *M. oleifera*, has in recent years attracted much attention for its antidiabetic properties in preclinical models (Gupta *et al.* 2012). Many parts of *Moringa* plant including leaves, immature pods, flowers and fruits are edible and are used as vegetables in several countries (Bennett *et al.* 2003). The aqueous extract of *M. oleifera* leaves has been reported to possess potent anti-hyperglycemic and anti-hyperlipidemic effects in both insulin-resistant and insulin-deficient rat models (Jaiswal *et al.* 2009). Similar benefits were also observed in diabetic patients with aqueous extract demonstrating effective hypoglycemic effect in humans (Kumar 2010; Ghiridhari *et al.* 2011). Earlier studies have been mostly carried out on *M. oleifera*, which is a species commonly found in the Indian subcontinent. In the present study, the efficacy of hydro-alcoholic extract of *M. peregrina* leaves (native to Africa and Arabian Peninsula) was evaluated for its antidiabetic potential using inhibition assays of enzymes considered to be effective targets for intervention in diabetes, such as α -amylase, α -glucosidase and dipeptidyl peptidase IV (DPP IV). The antioxidant property was also evaluated along with phytochemical analysis of the extract to identify major classes of phytochemicals which might be responsible for the observed pharmacological properties. In the context that *Moringa* is used in dietary cuisines and traditional medicine (Dieye *et al.* 2008; Robiansyah *et al.* 2014), the present evaluation aims to seek evidence in support of the importance of the plant as functional food for diabetes prevention. Such studies assume significance in the light of WHO (World Health Organization) observation that recognizes traditional medicine as “an accessible, affordable and culturally acceptable form of healthcare trusted by large numbers of people, which stands out as a way of coping with the relentless rise of chronic non-communicable diseases in the midst of soaring health-care costs and nearly universal austerity” (WHO 2013).

MATERIALS AND METHODS

Materials

Fresh leaves of *M. peregrina* were procured from the local market supplying plants popular for traditional and medicinal value (Tabuk, Saudi Arabia). These were authenticated by a plant taxonomist from the Department of Biology, Faculty of Science, University of Tabuk. Methanol and ultrapure water were purchased from Thermo Fisher Scientific (Fremont, CA). 2,2'-Azino-bis(3-ethylbenzothiazoline-6-sulfonic acid) diammonium salt (ABTS) was purchased from Fluka (St. Louis, MO). Enzymes α -amylase (EC 3.2.1.1) from porcine pancreas, α -glucosidase (EC 3.2.1.20) from rat intestine, dipeptidyl aminopeptidase (EC 3.4.14.5) IV from porcine kidney, Gly-Pro-p-nitroanilide hydrochloride and Phosphate buffered saline (PBS) were purchased from Sigma-Aldrich Co. (St. Louis, MO). All other chemicals and reagents were of analytical grade.

Extraction Procedure

The extraction procedure was followed as described by Suarez *et al.* (2010), with slight modifications. The *M. peregrina* leaves were rinsed with cold water and dried under shade. Dried plant material was crushed into small pieces in wooden mortar and pestle and further ground to fine dry powder with a blender. A quantity of 300 g dry leaves powder was then soaked in 2 L of 80% methanol in a conical flask placed in a water bath at 40°C for 24 h with constant shaking. The extracted mixture was then filtered through a double-layered clean cheesecloth and subsequently through double-layered Whatman paper (Sigma-Aldrich Co., St. Louis, MO). The filtrate was concentrated under reduced pressure at 35°C using a Buchi rotavapor R-210 (Flawil, Switzerland). Finally, the concentrated extract was further placed under vacuum at -30°C for 3–4 days to yield solid/thick pastes. The residual material was weighed to be 45 g giving a total yield of 15% (w/w).

Phytochemical Analysis

The extract was subjected to analysis for the presence of major classes of phytochemicals such as phenolics, tannins, alkaloids and saponins. Quantitative determination of total phenolic content was made as described earlier using pyrogallol as a standard compound (Blainski *et al.* 2013) and tannin was determined using tannic acid as a reference (Makkar 2000). The contents of alkaloids and saponins were determined by gravimetric estimation as described by Harborne (1976).

ABTS Radical Scavenging as Antioxidant Activity

Experiments were performed according to Re *et al.* (1999) with slight modifications. ABTS and potassium persulfate were dissolved in distilled water to a final concentration of 7 and 2.45 mM, respectively. These two solutions were mixed and the mixture was allowed to stand in the dark at room temperature for 16 h before use in order to produce ABTS radical (ABTS^{•+}). For the study of antioxidant capacity, the ABTS radical solution was diluted with distilled water to an absorbance of 1.00 at 734 nm. Extract (5–100 µg/mL) or gallic acid standards (final concentration 0.5–2.5 µg/mL) were added to diluted ABTS^{•+} solution (mixed) and the absorbance was recorded after 6 min at 734 nm using a microplate reader (Versamax, Molecular Devices, Sunnyvale, CA). Results were presented as % inhibition of radicals (compared to control without any test agent) and IC₅₀ was calculated using log-probit analysis.

α-Amylase Inhibition Assay

The α-amylase inhibitory activity for the *M. peregrina* extract was determined based on the spectrophotometric assay using acarbose as the reference compound (Gella *et al.* 1997). In brief, a pre-incubation mixture containing 250 mM phosphate buffer (pH 7.0) and acarbose (positive control)/extract of various concentrations along with 0.1 U of α-amylase enzyme was pre-incubated at 39°C for 15 min. Following pre-incubation, a substrate (CNP-G3) was added to a final concentration of 0.276 mM and the plate was incubated at 39°C for 60 min. The absorbance was measured at 405 nm in a microplate reader (Versamax). A control reaction was carried out without the test sample. Results were presented as % inhibition of enzyme activity (compared to control without any test agent) and IC₅₀ was calculated using log-probit analysis.

α-Glucosidase Inhibition Assay

The assay was performed as described by Vogel and Vogel (1997) with slight modifications. In brief, the pre-incubation mixture contained 80 mM phosphate buffer (pH 7.0) and acarbose (positive control)/extracts of various concentrations along with 0.1 U of α-glucosidase enzyme. The reaction mixture was mixed and pre-incubated at 37°C for 30 min. Following pre-incubation, a substrate (sucrose) was added to a final concentration of 23.125 mM and the reaction mixture was mixed and incubated at 37°C for 50 min. The reaction was stopped by keeping in boiling water bath for 2 min and then cooled to room temperature. Up to 250 µL of glucose reagent was added to 50 µL of reaction mixture, mixed and incubated at 25°C for 10 min. The

absorbance was measured at 510 nm in a microplate reader (Versamax). A control reaction was carried out without the test sample. Results were presented as % inhibition of enzyme activity (compared to control without any test agent) and IC₅₀ was calculated using log-probit analysis.

DPP IV Inhibition Assay

The assay was carried out as per the method of Kojima *et al.* (1980) with some modifications. In brief, the pre-incubation mixture contained 100 mM Tris-HCl buffer (pH 8.2) and Ile-Pro-Ile (positive control)/extract of various concentrations along with 0.3 U of DPP enzyme. The plate was mixed and pre-incubated at 37°C for 10 min. Following pre-incubation, substrate (Gly-Pro-p-nitroanilide hydrochloride) was added to a final concentration of 0.73 mM and mixed. The plate was incubated at 37°C for 30 min. The absorbance was measured at 405 nm in a microplate reader (Pherastar, BMG Labtech, Cary, NC). Controls were also run which were devoid of test samples. Results were presented as % inhibition of enzyme activity (compared to control without any test agent) and IC₅₀ was calculated using log-probit analysis.

Statistical Analysis

All experiments were performed in three different sets, with each set in triplicate. The data are expressed as mean ± SEM (standard error of the mean). Statistical analysis was performed for ANOVA (analysis of variance) followed by *F*-test using SPSS version 11.5 (SPSS, Inc., Chicago, IL). Values of *P*, which were ≤0.05, were considered significant.

RESULTS AND DISCUSSION

M. peregrina Extract Exhibits the Presence of Bioactive Phytochemicals

The quantitative analysis of the hydro-alcoholic extract of *M. peregrina* showed the presence of major classes of phytochemicals which are considered to possess wide range of pharmacological properties (Ramawat *et al.* 2009). The quantification was made for four kinds of molecules, and as shown in Fig. 1 the extract contained phenolics, tannins, alkaloids and saponins in varying concentrations. The highest concentration was recorded for the saponins compared to other classes of phytochemicals. Studies have demonstrated the antidiabetic properties of these phytochemicals in preclinical models through mechanisms that include stimulation of insulin secretion and enhance uptake of glucose by tissues, inhibition of glucose absorption from intestine, retardation of glucose production from liver and improved β-cell function (Ahmed *et al.* 2010). In a

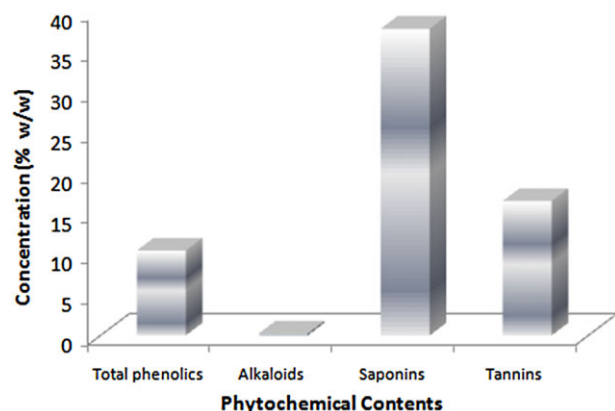


FIG. 1. PHYTOCHEMICAL CONTENT OF *MORINGA PEREGRINA* EXTRACT EXAMINED FOR SELECTED CLASSES OF BIOACTIVE MOLECULES INCLUDING PHENOLICS, ALKALOIDS, SAPONINS AND TANNINS

recent study, saponins have been shown to lower blood glucose levels by stimulation of insulin secretion and enrichment of β -cell density in animal model (Koneri *et al.* 2014). Potentials of tannins derived from plant sources have also been evaluated with studies demonstrating anti-diabetic properties through inhibition of α -amylase and α -glucosidase activities (McDougall *et al.* 2005; Barrett *et al.* 2013). It is thus presumed that the pharmacological properties of *M. peregrina* might be associated with the significant presence of saponins and tannins serving as bioactive nutraceuticals.

Inhibition Profile of ABTS Radicals by *M. peregrina* Extract Shows a Dose-Dependent Curve

The imbalance of cellular redox homeostasis, which corresponds to the equilibrium of reactive oxygen species and the cellular antioxidant system, contributes to the pathogenesis of most of the chronic diseases including diabetes (Wright *et al.* 2006). Therefore, for most of these chronic diseases associated with poor dietary habits, the major prophylactic strategy includes regular consumption of fruits and vegetables rich in antioxidant phytochemicals. *Moringa* is a part of human diet among the population of many countries (Bennett *et al.* 2003). In this study, the antioxidant capacity of *M. peregrina* extract was evaluated using ABTS radical scavenging assay (Re *et al.* 1999). As shown in Fig. 2, the *M. peregrina* extract led to the progressive scavenging of ABTS $^{\bullet+}$ radical in a dose-dependent manner with an IC_{50} value recorded at 20.56 μ g/mL. These results correlate well with the significant concentration of saponins and moderate level of phenols in the extract as these molecules have been shown to have antioxidative properties (Sabu and

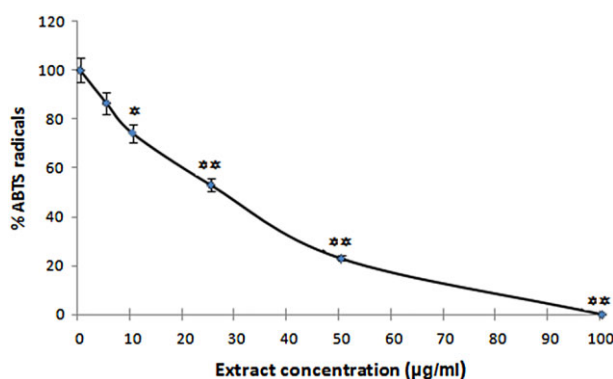


FIG. 2. RADICAL SCAVENGING CAPACITY OF *MORINGA PEREGRINA* EXTRACT ASSESSED BY ITS ABILITY TO INHIBIT ABTS $^{\bullet+}$ RADICALS. Values reported are $\pm$ SEM of three independent experiments. * $P \leq 0.05$; ** $P \leq 0.01$; significant when compared with control (in the absence of tested concentrations of extract).

Kuttan 2002; Chen *et al.* 2014). Diabetes mellitus is a disease with evidence of oxidative stress as an etiological factor; therefore, it is believed that antidiabetic pharmacological may also contain anti-peroxidative/cytoprotective potentials (Sabu and Kuttan 2002; Wright *et al.* 2006).

Inhibitory Effect of *M. peregrina* Extract on Mammalian Pancreatic α -Amylase

Pancreatic α -amylase is a critical enzyme in the digestive pathway of carbohydrates and catalyzes the initial step in hydrolysis of starch to a mixture of smaller oligosaccharides consisting of maltose, maltotriose, and a number of α -(1-6) and α -(1-4) oligoglucans. Degradation of this dietary starch proceeds rapidly and leads to elevated postprandial hyperglycemia. It has been shown that the activity of human pancreatic α -amylase in the small intestine correlates with an increase in postprandial glucose levels, the inhibition of which is therefore an important intervention in management of type 2 diabetes (Tarling *et al.* 2008). In the experiment demonstrated in Fig. 3, α -amylase activity was examined in the presence of varying concentration of *M. peregrina* extract. The result showed consistent interference of the extract with the activity of the enzyme in response to increasing concentration of the extract with IC_{50} value of 1,335.89 μ g/mL. Inhibitors of pancreatic α -amylase retard carbohydrate digestion leading to a reduction in the rate of glucose absorption and lowering the postprandial serum glucose levels (Cheng and Fantus 2005).

Inhibitory Effect of *M. peregrina* Extract on Mammalian Intestinal α -Glucosidase

Mammalian α -glucosidase anchored in the mucosal brush border of the small intestine catalyzes the end step digestion

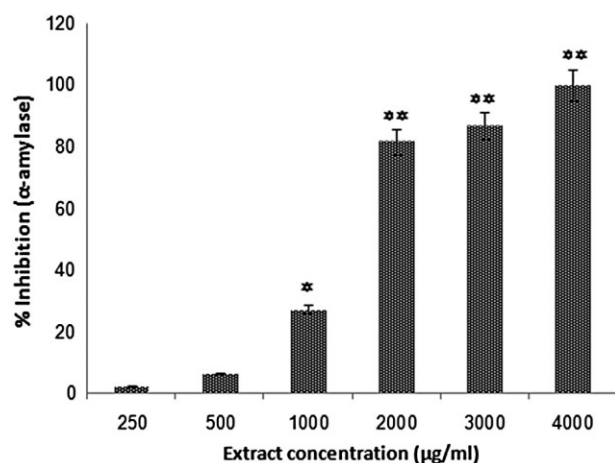


FIG. 3. INHIBITION PROFILE OF MAMMALIAN α -AMYLASE BY THE EXTRACT OF *MORINGA PEREGRINA*

Values reported are $\pm$ SEM of three independent experiments.

* $P \leq 0.05$; ** $P \leq 0.01$; significant when compared with control (in the absence of tested concentrations of extract).

of oligosaccharides and sucrose that are abundant carbohydrates in human diet (Van de Laar 2008). Inhibitors of α -glucosidase retard the breakdown of carbohydrate in small intestine and diminish the postprandial blood glucose excursion in diabetic subjects in a manner similar to the upstream inhibition of α -amylase (Van de Laar *et al.* 2005) and thus have a subsiding effect on postprandial blood glucose and insulin levels. The α -glucosidase inhibition potential of the extract was measured as shown in Fig. 4. It displayed moderate inhibition of α -glucosidase by the extract with IC_{50} value of 3,256.68 μ g/mL. Previous studies

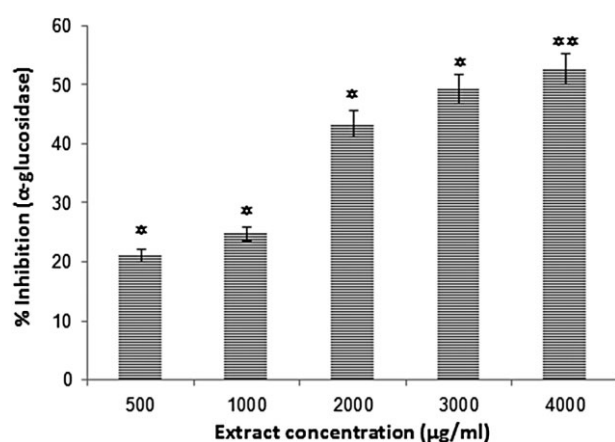


FIG. 4. INHIBITION PROFILE OF MAMMALIAN α -GLUCOSIDASE BY THE EXTRACT OF *MORINGA PEREGRINA*

Values reported are $\pm$ SEM of three independent experiments.

* $P \leq 0.05$; ** $P \leq 0.01$; significant when compared with control (in the absence of tested concentrations of extract).

on α -glucosidase inhibitors isolated from medicinal plants suggest that several potential inhibitors belong to saponin and tannin classes which possess the characteristic structural features to inhibit α -glucosidase enzyme (Xiao *et al.* 2013; Lee *et al.* 2014).

Progressive Inhibition of Mammalian DPP IV by *M. peregrina* Extract

Results from DPP IV inhibition assay (Fig. 5) revealed that the extract of *M. peregrina* progressively inhibited the enzyme activity in a dose-dependent manner with an IC_{50} value of 1,218.12 μ g/mL. DPP IV inhibition results in advancing the half-life and biological activity of incretin hormones glucagon-like peptide-1 (GLP-I) and glucose-dependent insulintropic polypeptide (GIP) (Drucker and Nauck 2006). GLP-I and GIP under normal metabolic conditions improve glucose homeostasis by promoting α - and β -cell functions, insulin secretion, and glycogenesis in muscles and liver (Yabe and Seino 2011). Furthermore, these also down-regulate the level of gastric emptying and gastric acid secretion to reduce postprandial glucose uptake (Drucker and Nauck 2006). However, GLP-1 and GIP have short plasma half-life of 1–2 min due to rapid inactivation by DPP IV enzyme (Reimer *et al.* 2002). Inhibition of DPP IV has been regarded as an efficient strategy to manage type 2 diabetes and several inhibitors commercialized as drugs are currently in use such as sitagliptin and vildagliptin (Lyseng-Williamson 2007; Croxtal and Keam 2008). Sulfonylurea acts by increasing the insulin secretion in a glucose-independent manner, thereby risking severe unpredictable

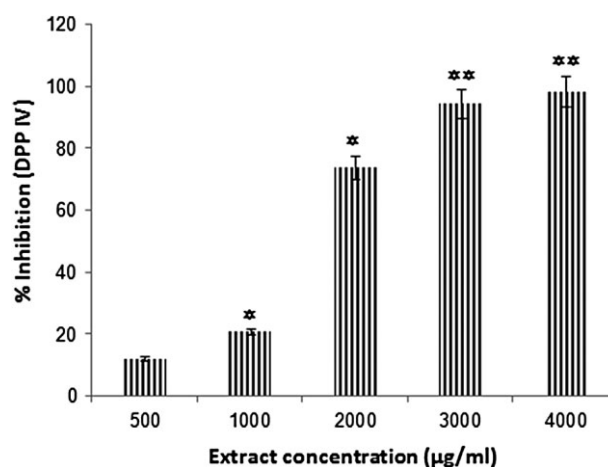


FIG. 5. INHIBITION PROFILE OF MAMMALIAN DPP IV BY THE EXTRACT OF *MORINGA PEREGRINA*

Values reported are $\pm$ SEM of three independent experiments.

* $P \leq 0.05$; ** $P \leq 0.01$; significant when compared with control (in the absence of tested concentrations of extract).

hypoglycemia, particularly if the meal is delayed or its carbohydrate quantity reduced, whereas metformin has been found to be associated with adverse gastrointestinal effects. Therefore, introduction of DPP IV inhibitors appeared to be useful as these can augment the incretin system, thereby promoting β -cell efficiency and insulin release in a glucose-dependent manner (Dicker 2011; Gupta and Kalra 2011). The potential of *M. peregrina* extract to inhibit DPP IV enzyme provides a promising approach of functional foods as alternative medicine to limit the adverse effects associated with pure and synthetic drugs in prevention of diabetes.

CONCLUSION

Our laboratory focuses on engaging dietary factors in the prevention of chronic diseases including diabetes and cancer (Ullah *et al.* 2014a,b). The results provide mechanistic insight toward evidence-based support for use of *Moringa* species as functional food in traditional medicine for diabetes management in Asian-African countries (Dieye *et al.* 2008). The efficiency of *M. peregrina* in the inhibition of α -amylase and α -glucosidase enzymes obtained in our study is comparable to other species such as *M. oleifera* and *M. stenopetala* in which *M. peregrina* demonstrated higher activity against α -amylase but lower activity against α -glucosidase compared to other two species (Adisakwattana and Chanathong 2011; Toma *et al.* 2014). However, there are no reports to show *M. oleifera* and *M. stenopetala* as DPP IV inhibitors unlike *M. peregrina* as reported in this study. This is the first mechanism-based study of *M. peregrina* (native to Arabian Peninsula) that accounts for the hypoglycemic effect of the plant observed in preclinical models (El-Alfy *et al.* 2011). An animal-based study had previously shown that the aqueous and alcoholic extracts of aerial parts of *M. peregrina* significantly decreased the blood glucose level of streptozotocin diabetic rats (El-Alfy *et al.* 2011). Thus, our study complements the previous observations in animal model and traditional medicine in terms of highlighting the putative mechanism for the antidiabetic properties of *M. peregrina* observed *in vivo*. The plant displays two distinct advantages in its pharmacological potential against diabetes: (1) the extract of *M. peregrina* inhibits the intestinal glycosidases (α -amylase and α -glucosidase) at a mild-to-moderate level that could induce a balanced postprandial hypoglycemia with limited gastrointestinal adversity caused by undigested carbohydrates, compared to conventional drugs inhibiting these enzymes completely; (2) the extract efficiently inhibits DPP IV enzyme and thus has advantages of DPP IV inhibitors that promote β -cell efficiency and long-term efficacy through regulation of incretin hormones and glucose-dependent release of insulin. Moreover, because *Moringa* is being used by

humans in different dietary forms, it has a higher index of safety. Thus, the regular consumption of *M. peregrina* might be potentially beneficial as prophylactic intervention for populations at borderline diabetic conditions as well as for alleviating transient hyperglycemic conditions. However, controlled trials on human subjects and pharmacokinetic studies would be essential to establish the dose–effect relationship.

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CONFLICT OF INTEREST STATEMENT

The authors confirm that this article content has no conflicts of interest. The authors alone are responsible for the content and writing of the paper.

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